Hardware Security

硬體安全

Power Consumptions of Cryptographic Devices

Wei-Chih Hong

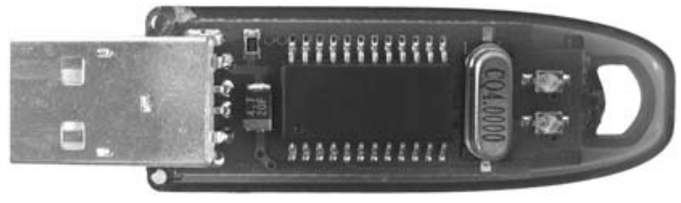
Outline

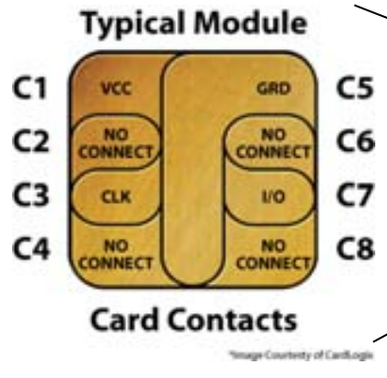
- Cryptographic devices
- IC design and implementation
- Power consumption

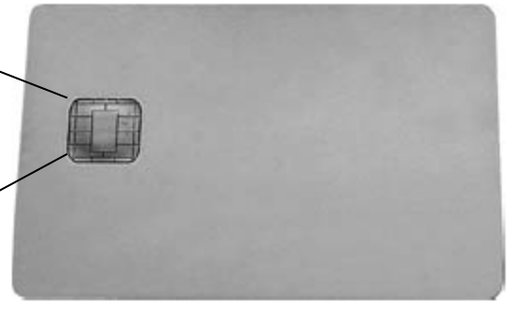
Cryptographic Devices

- **Operational components**
  - Dedicated hardware
  - General purpose hardware
  - Cryptographic software
- **Data handling components**
  - Memory
  - Interface









Smart Card

Smart Cards



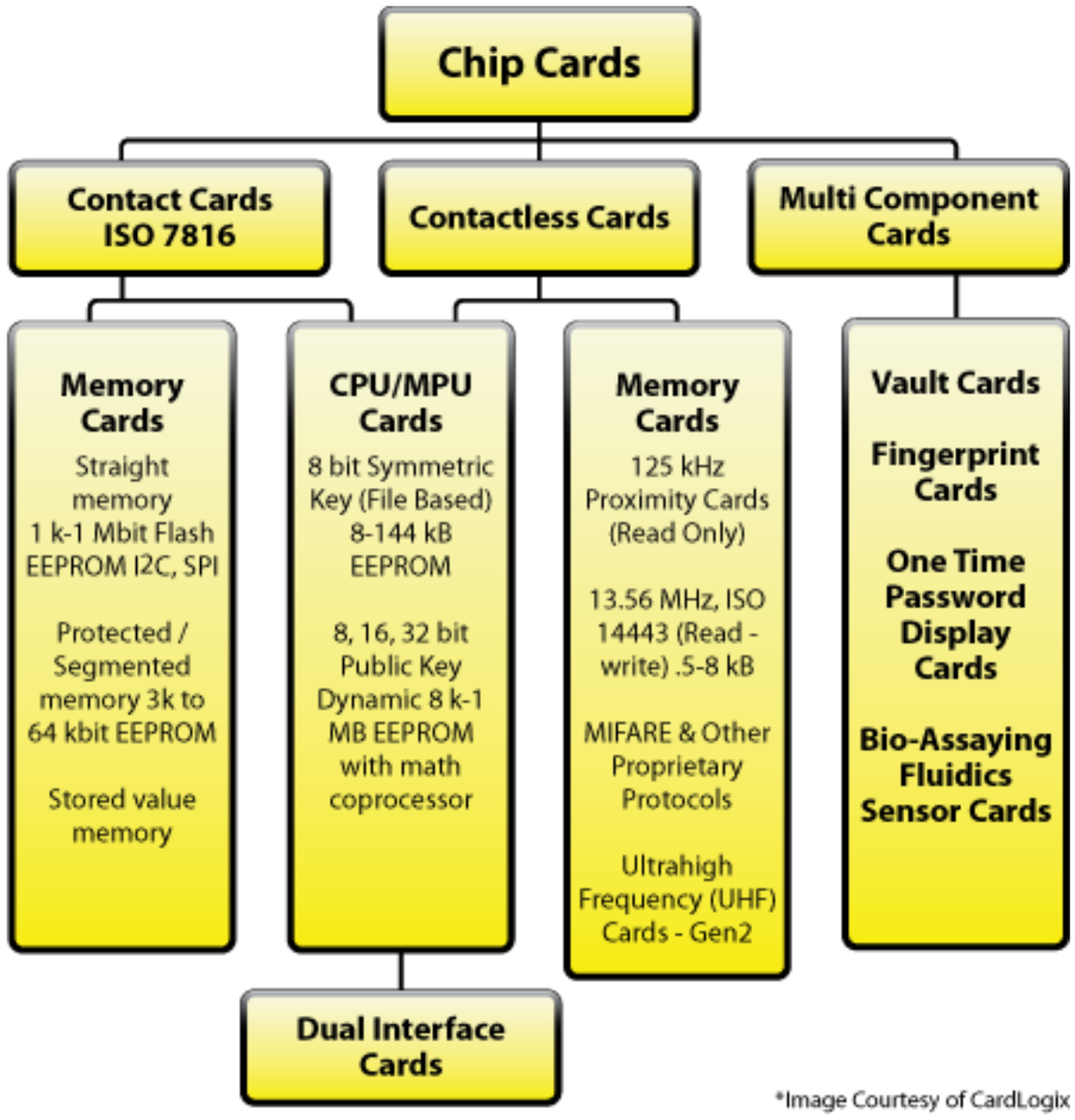


*Image Courtesy of CardLogix

http://www.smartcardbasics.com/smart-card-types.html

Why Integrated Circuits?



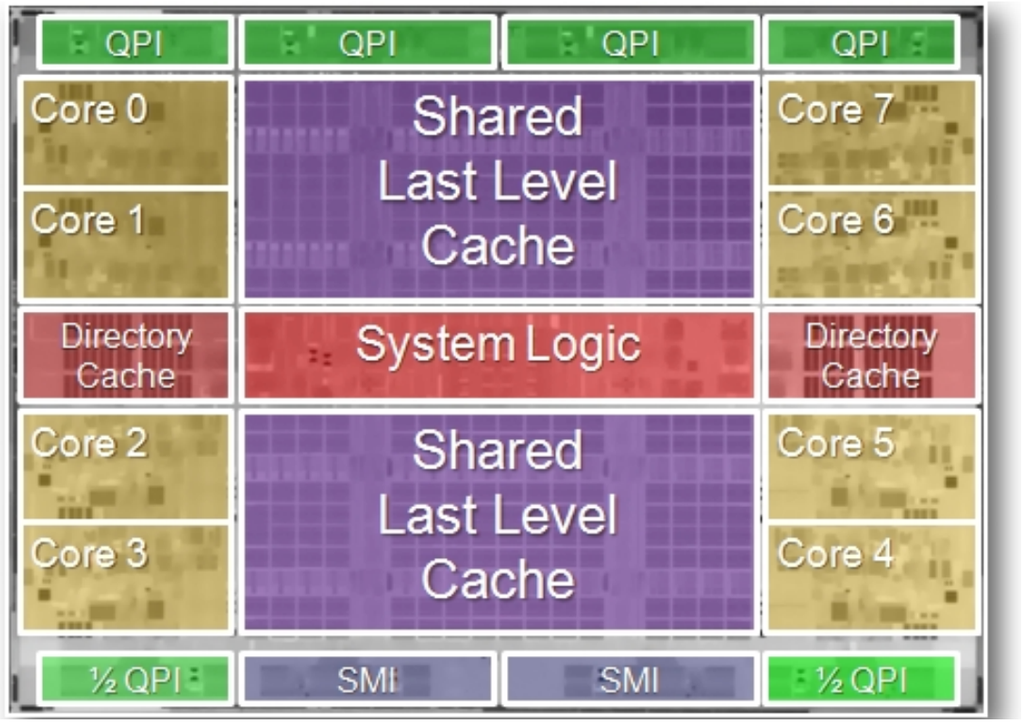
ENIAC (1946)

World's first general purpose processor

17 thousand vacuum tubes

Frequency: 5 KHz

Area: 67 m^2; Power: 150 KW



Intel Itanium (2011)

32-nm 8-core 64-bit processor

3.1 billion transistors

Frequency: 2 GHz

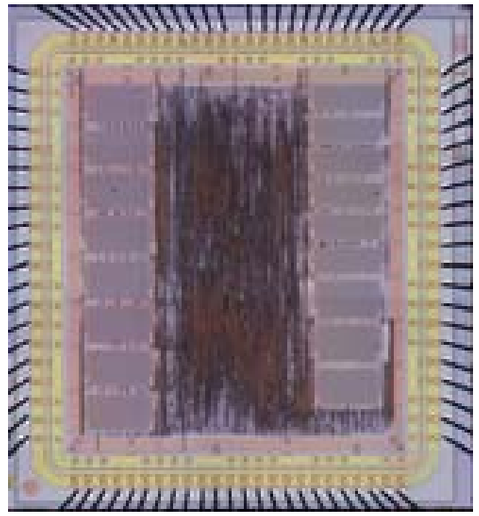
Area: 544 mm^2; Power: 170 W

Small, but Extremely Complex

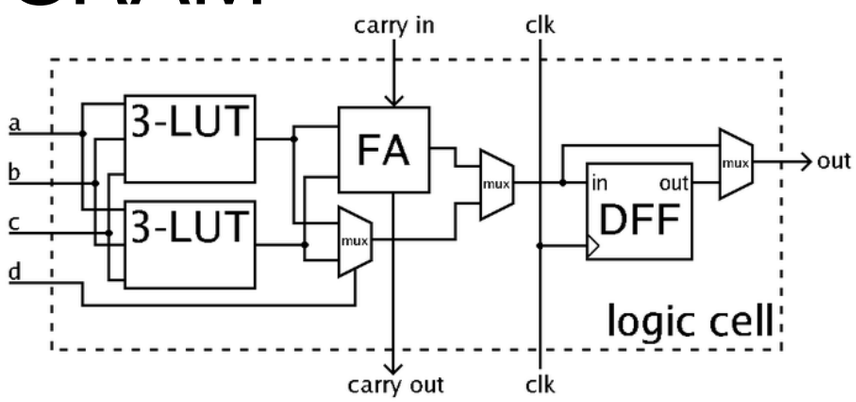
- **Chip design**
  - Hundreds of people may work for one project
- **Tapeout and wafer fabrication**
  - Hundreds of steps of process may be applied
- **Testing and assembly**
  - Testing time v.s. IC yield

Design and Implementations

- **ASIC (Application-specific integrated circuit)**
  - Full custom (analog, RF circuits)
  - Semi custom (CPU)
  - Cell-based



- **FPGA (Field-programmable gate array)**
  - Embedded blocks, e.g. CPU, SRAM
  - LUT-based cells

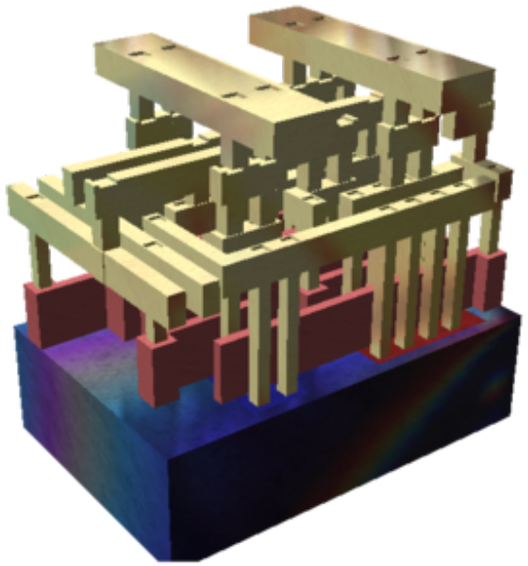


Cell-based Methodology

- Algorithms described using HDL (hardware description language)
- Layout generated by CAD tools
- **Achieving complex digital ASICs efficiently**
  - with the help of abstraction hierarchy

Standard Cell Library

- **A standard cell is a group of transistors and interconnection for low-level logic functions**
  - Boolean functions (e.g. AND, OR, XOR…)
    - combinational circuits
  - Storage functions (e.g. flip-flop)
    - sequential circuits



- **Usually optimized for different purposes (low power or high performance) and calibrated for different technologies**

Cell-based Design Flow

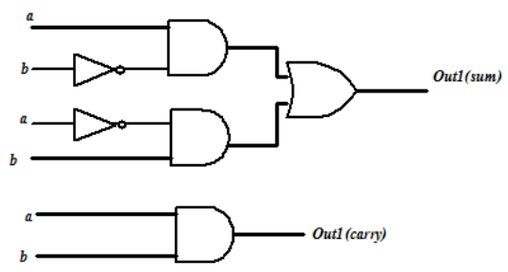
HDL

```
module chip_top(clk,

input clk, srt_n;
input [7:0] in_a;
output [7:0] out_b;
...
```

Behavioral design

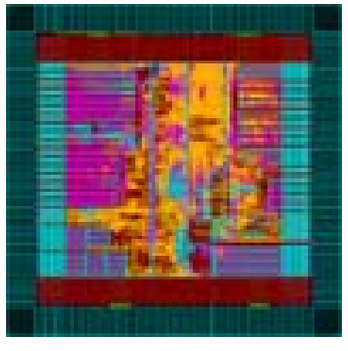
Logic synthesis



Structural design

Physical design

Layout



Design iterations

High-level design capture

Design synthesis

Floorplanning

Placement

Routing

Tape-out

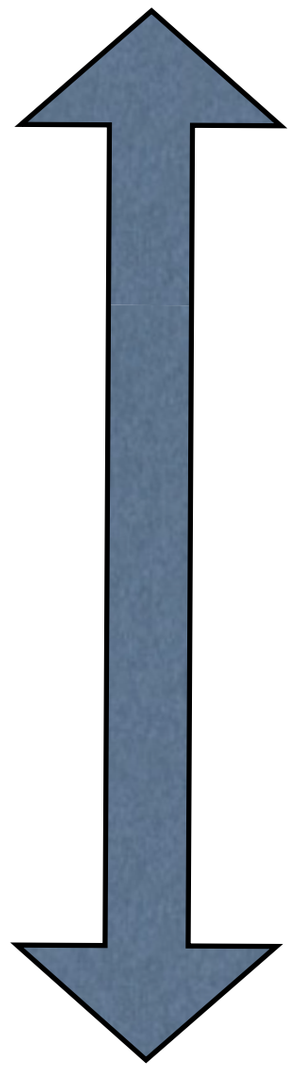
Front-end verification

Back-end verification

Multiple Levels of Abstraction

- **Behavioral Level**
  - algorithms
- **Register Transfer Level (RTL)**
  - net and registers
- **Gate Level**
  - logic gate primitives
- **Switch Level**
  - switch (physical primitives)

Less detail
- faster design entry
- faster simulation

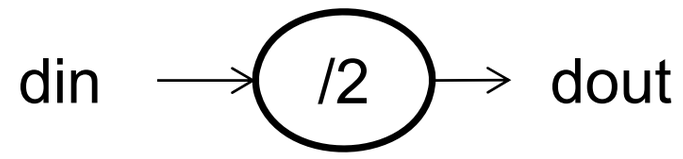


More detail
- slower design entry
- slower simulation

Abstraction Level Example

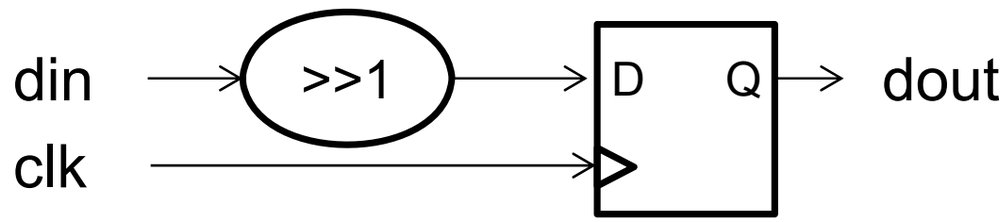
- **Behavioral**

```
always@(din)
  dout = din/2;
```



- **RTL**

```
always@(posedge clk)
  dout <= din>>1;
```

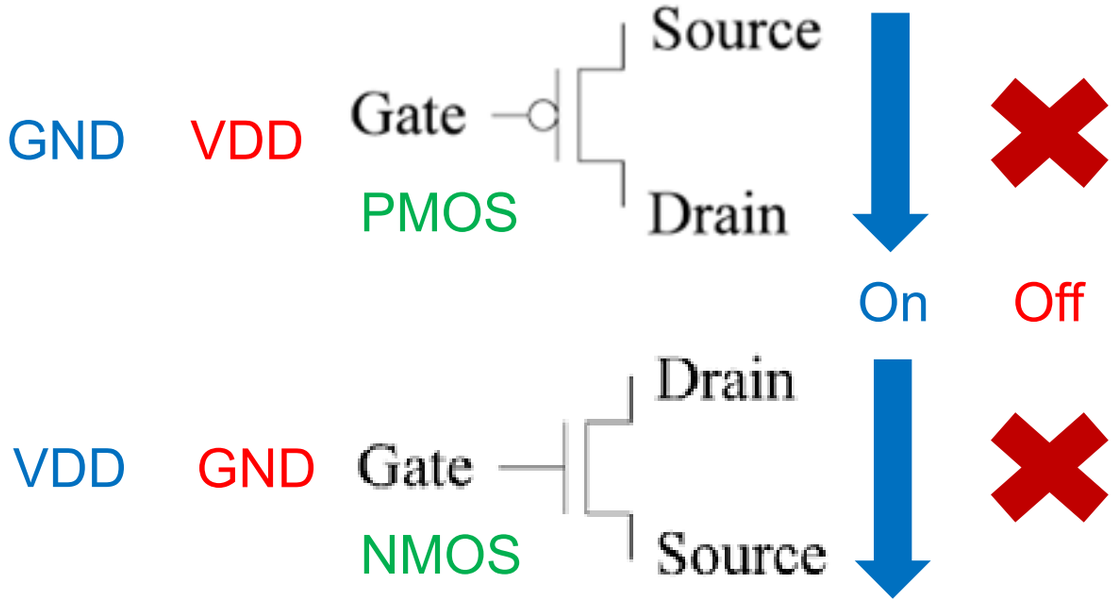


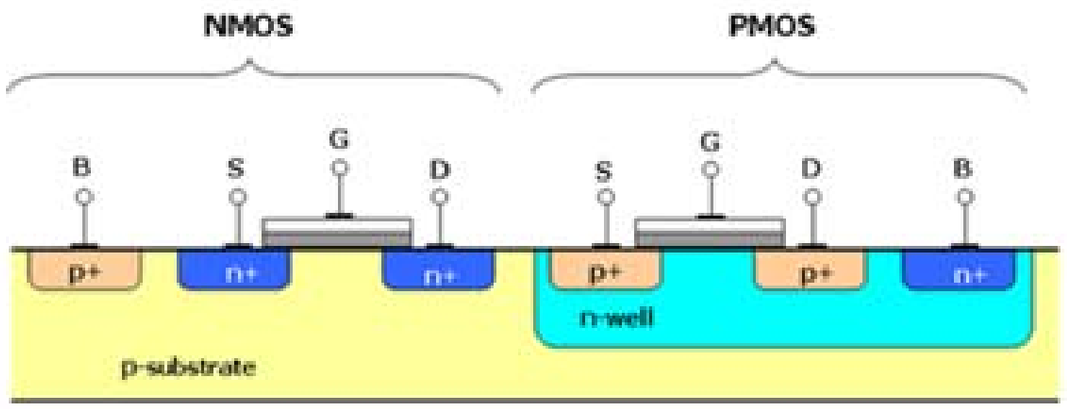
- **Gate**

```
DFF D0 (.D(din[1]), .Q(dout[0]), .CK(clk));
DFF D1 (.D(din[2]), .Q(dout[1]), .CK(clk));
DFF D2 (.D(din[3]), .Q(dout[2]), .CK(clk));
DFF D3 (.D(1'b0),   .Q(dout[3]), .CK(clk));
```

CMOS Technology

- **Complementary Metal-Oxide-Semiconductor**
  - Most popular IC technology
  - Voltage-controlled logic cells
  - Threshold voltage V_t



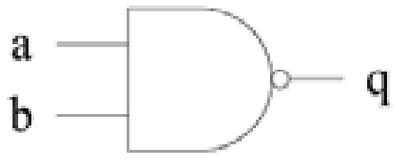


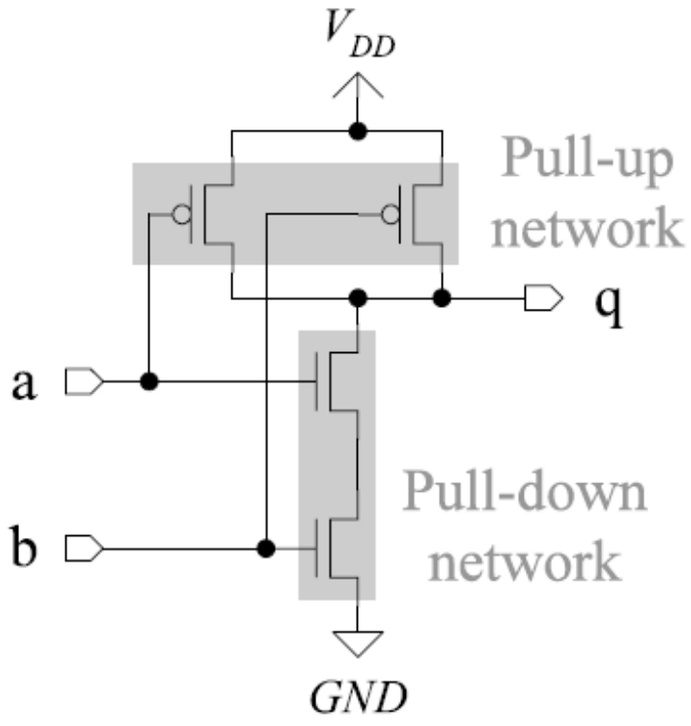
Cross Section

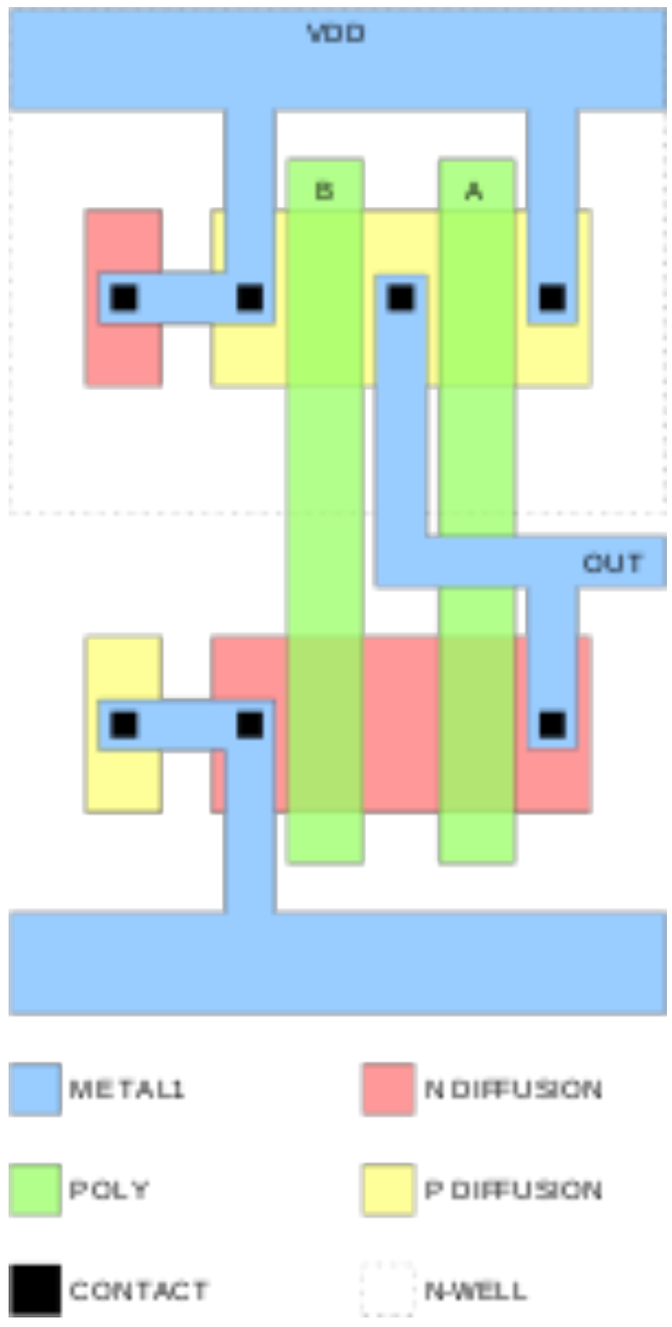
CMOS Logic Cells

- Low static power consumption
  - Leakage current
- Duality

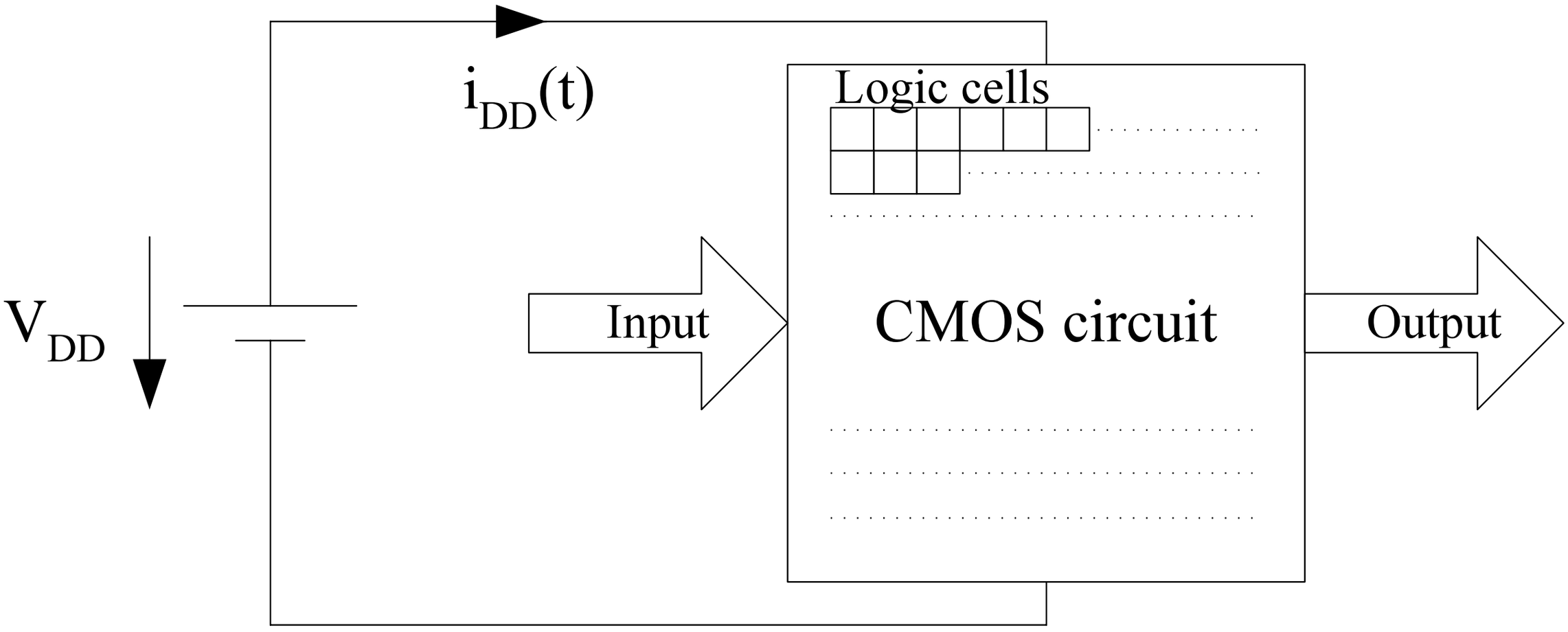
$$q = \overline{a \cdot b}$$







Power Consumption of CMOS Circuits



$$P_{cir} = \frac{1}{T}\int_0^T p_{cir}(t)dt = \frac{V_{DD}}{T}\int_0^T i_{DD}(t)dt$$

Power Consumption of CMOS Circuits

- Static power consumption
  - Leakage current (leakage power)
- Dynamic power consumption
  - Charging current (switching power)
  - Short-circuit current (internal power)

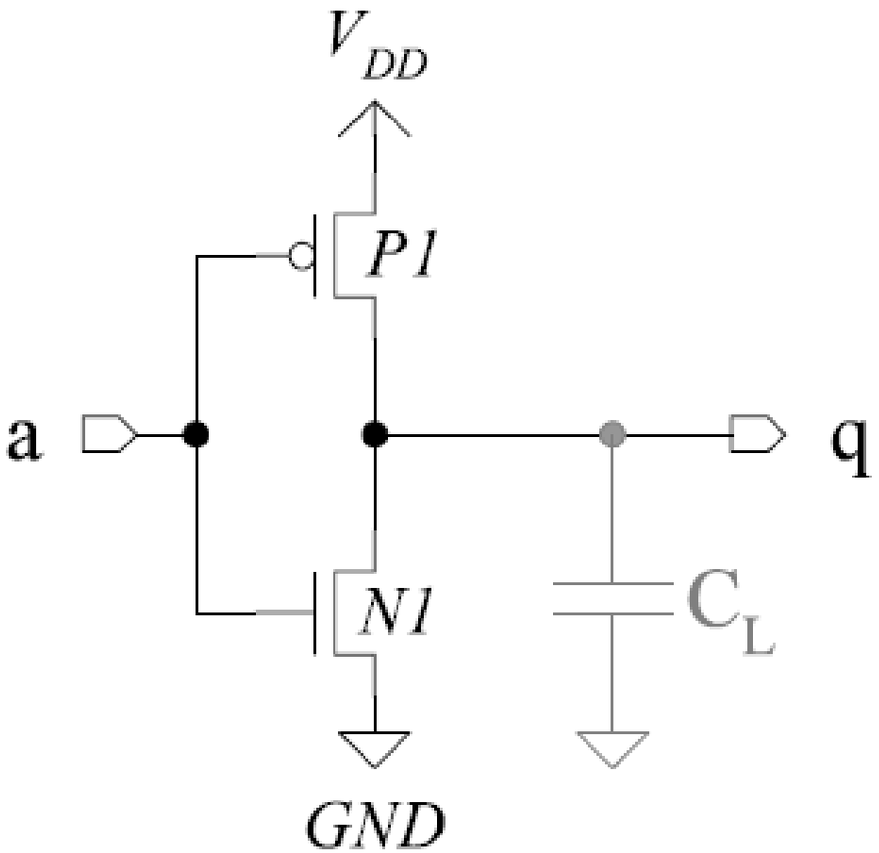
Leakage Power

- **Leakage current**
  - Sub-threshold conduction (source-to-drain)
  - Tunneling current (through gate oxide)

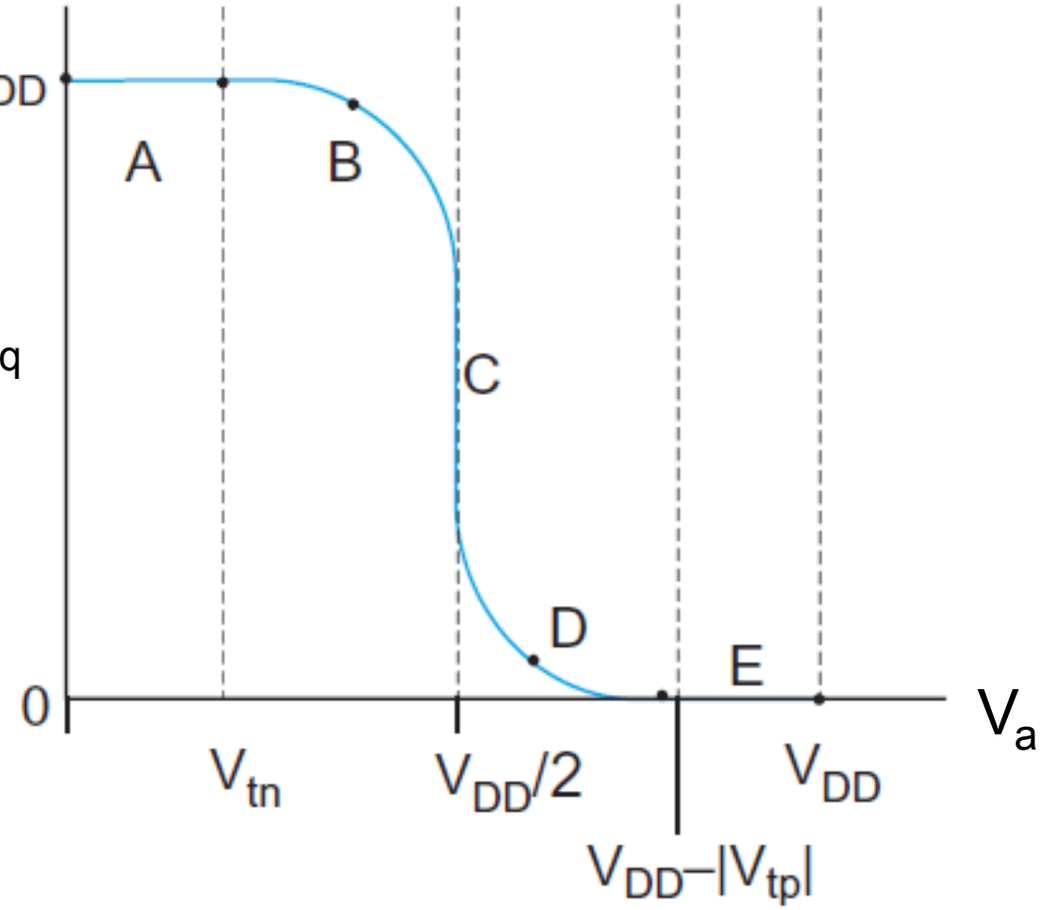
$$P_{stat} = I_{leak} \cdot V_{DD}$$

- **Small for early technologies, but significant for modern process technologies**
  - Can be alleviated by raising threshold voltage, but may increase dynamic power or slow down the circuits

Gate Switching



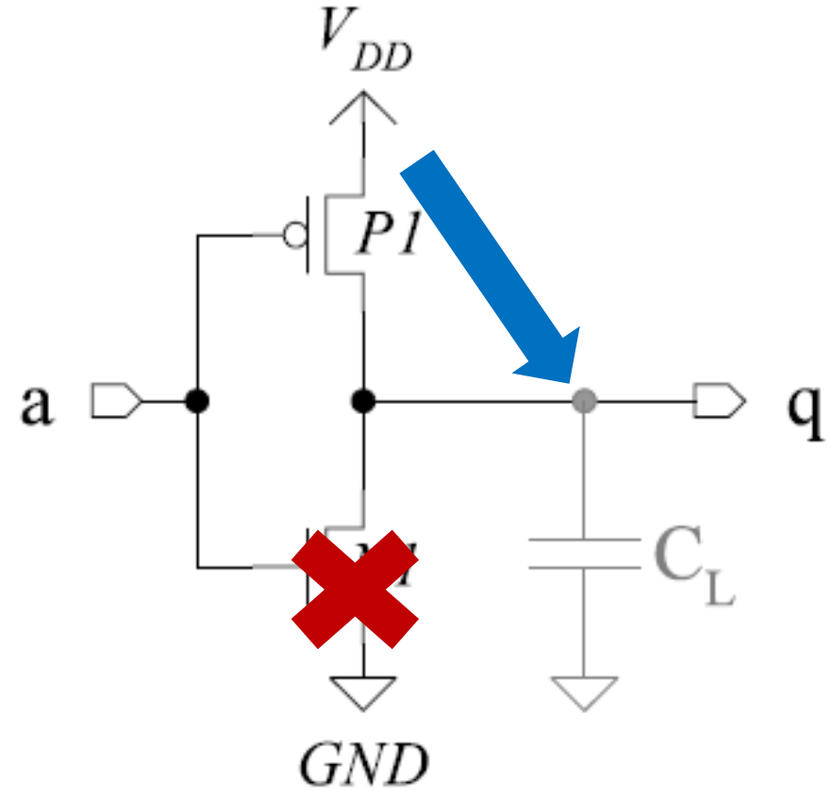
DC Transfer Characteristic



Switching Power

- Charging current
- Example: Lumped-C model

1->0



$$V_q = \frac{Q}{C_L}, I = \frac{dQ}{dt} = C_L \frac{dV_q}{dt}$$

Stored energy (in C_L):

$$\int V_q I dt = \int_0^{V_{DD}} C_L V_q dV_q = \frac{1}{2} C_L {V_{DD}}^2$$

Consumed energy (from V_{DD}):

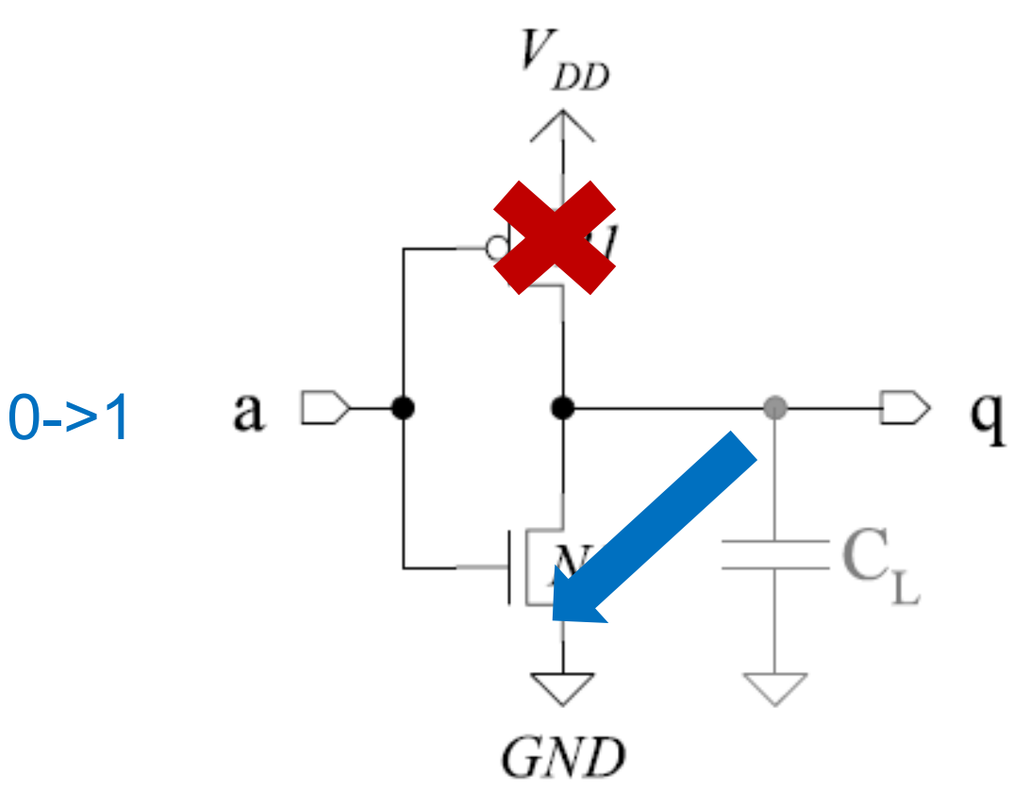
$$\int V_{DD} I dt = \int_0^{V_{DD}} C_L V_{DD} dV_q = C_L {V_{DD}}^2$$

Dissipated energy (Mostly in P1):

$$\frac{1}{2} C_L {V_{DD}}^2$$

Switching Power

- Charging current
- Example: Lumped-C model



Dissipate the stored energy and doesn't consume power from VDD

One full transition 0->1->0 consumes energy $C_L {V_{DD}}^2$

$$P_{chrg} = \frac{1}{T}\int_0^T p_{chrg}(t)dt = \alpha \cdot f \cdot C_L \cdot V_{DD}^2$$

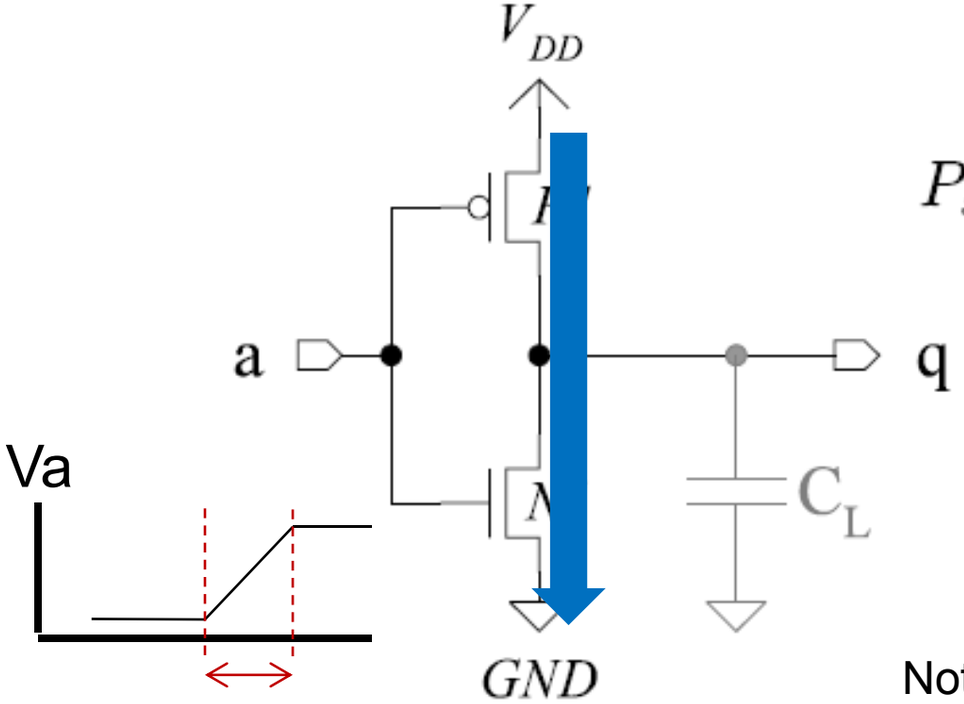
f: Clock frequency

α: activity factor

(average number of q 0->1 transition per cycle)

Internal Power

- **Short-circuit current**
  - Direct current from VDD to GND when NMOS and PMOS are both turned on.

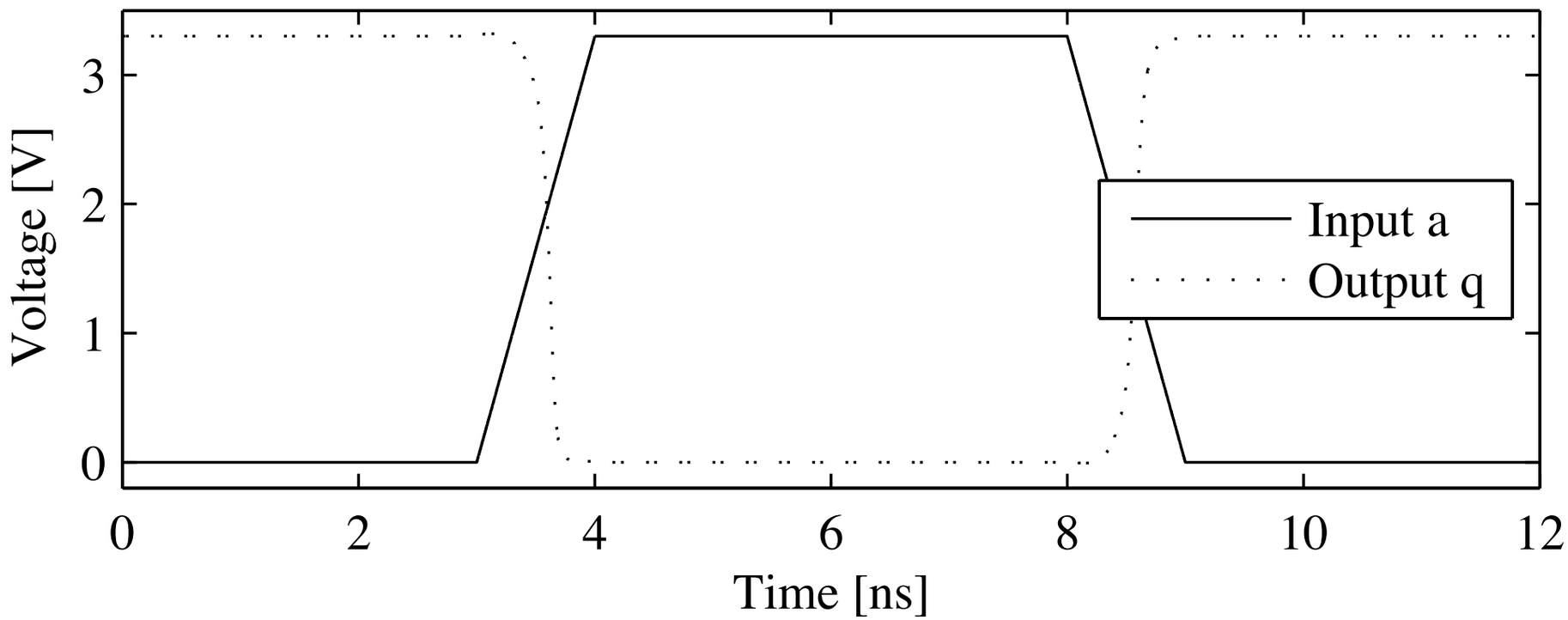


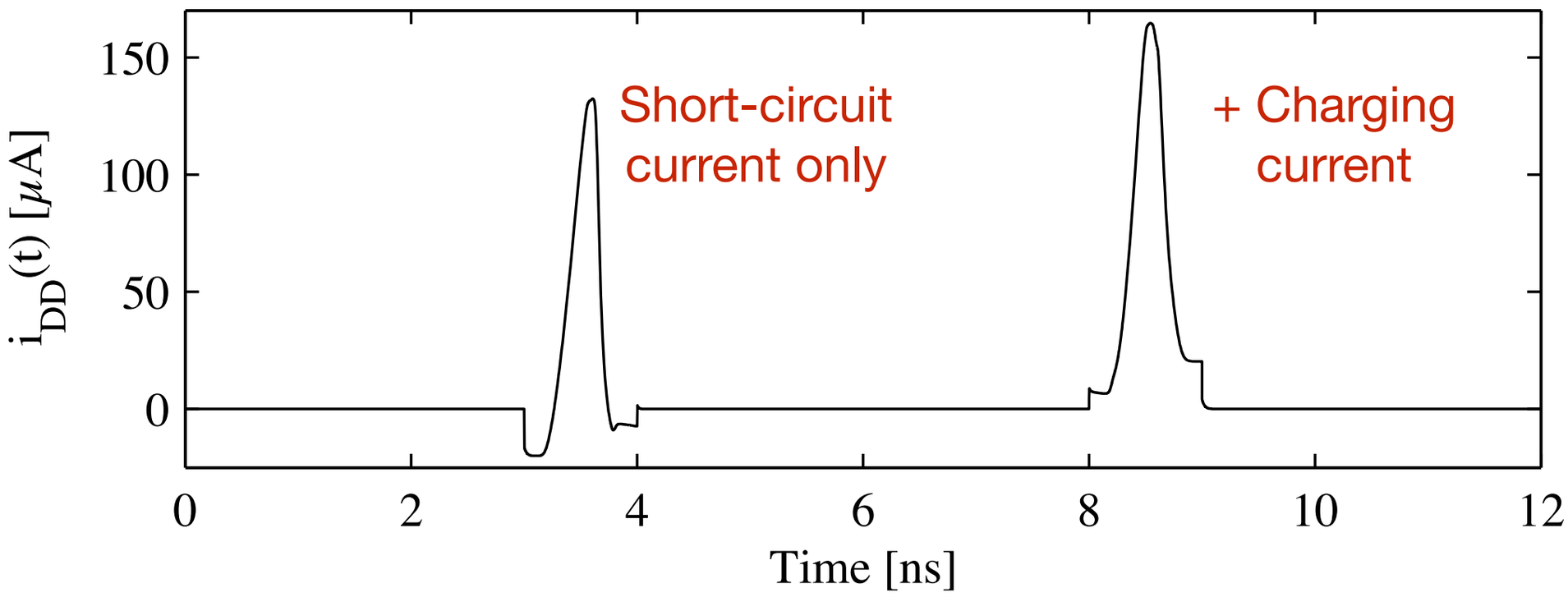
Transition Time

$$P_{sc} = \frac{1}{T}\int_0^T p_{sc}(t)dt = \alpha \cdot f \cdot V_{DD} \cdot I_{peak} \cdot t_{sc}$$

Note: For a common standard cell library,
input transition time and output capacitance are
the two inputs of calibrated look-up tables for power and delay.

Simulation of Dynamic Power

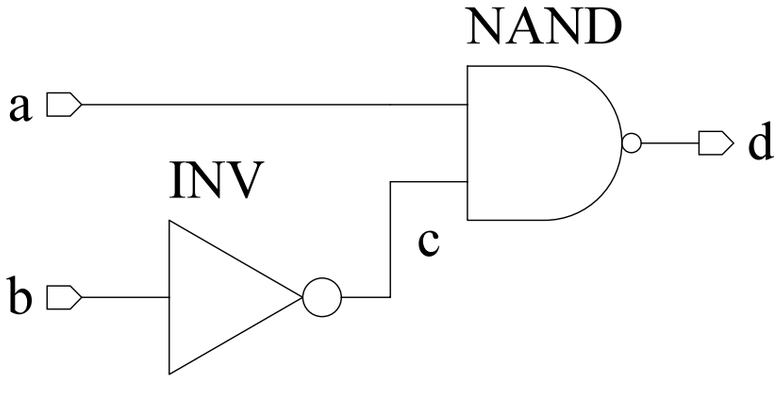


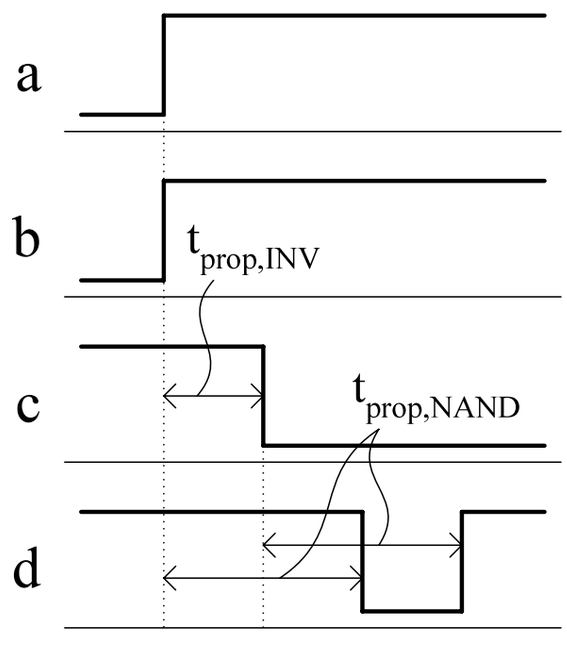


How about i_{GND}?

Glitches

- Unwanted transitions when combinational circuits propagate their values
  - different propagation delays
  - causing increased power consumption and delay
- Dependent on input patterns as well as operating conditions





Power Model for Designers

- Designers can have access to detailed information, such as netlist and layout, so the power simulation can be done very accurately.

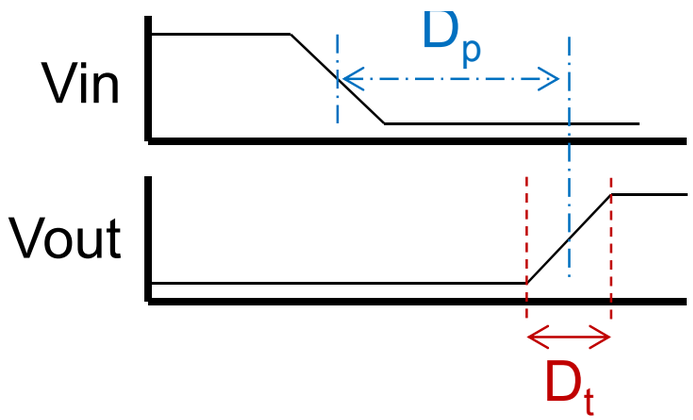
  - Analog level (transistor + precise RC)
  - Netlist level (standard cell + back-annotated timing)
  - Logic level (logic cell + transition count)

Netlist Simulation

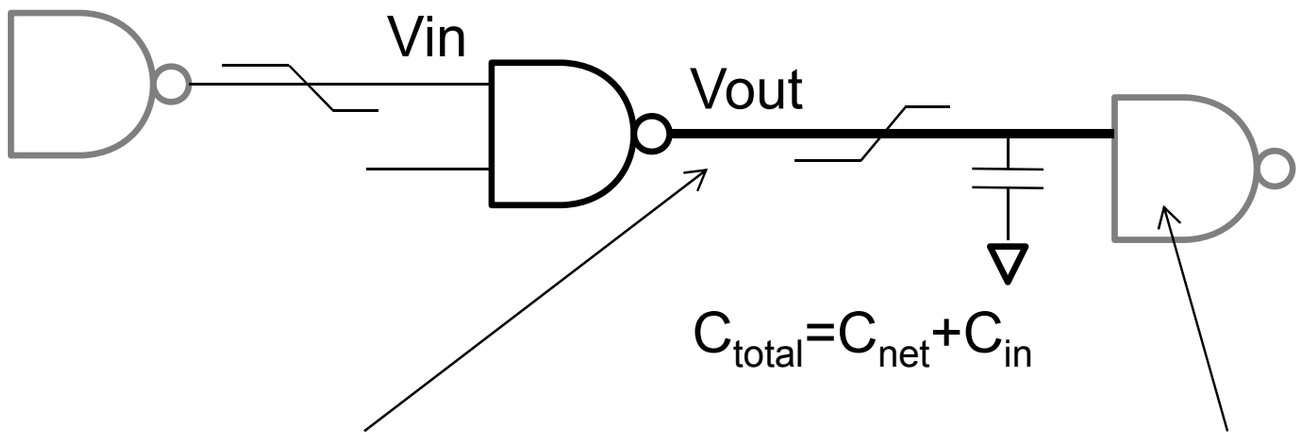
- **Gate-level Netlist**
  - Post-synthesis netlist
  - Post-layout netlist
    - accurate RC extraction from layout
- **Delay calculation**
  - Based on calibrated lookup table, the delay and power of different cases for each cell are derived
- **Dynamic simulation**
  - Back-annotate the delay information
  - Power consumption calculated by simulated waveform

Delay Model for Netlist Simulation

- Usually calibrated and approximated by tables for *propagation delay* D_p and *transition time* D_t



D_p and D_t both 2-D functions (tables) of **input transition time** $D_{t,in}$ of Vin and **output total capacitance** C_{total}



Wire has parasitic C_{net}, R_{net} (estimated by models)

Cell has input capacitance C_{in} (accurate calibration)

Logic-level Simulation

- **Gate-level netlist**
  - Only needs the function of gates
- **Hamming-distance model**
  - Considers the transition of each signal
  - Same weight for 0->1 and 1->0 $HD(v_0, v_1) = HW(v_0 \oplus v_1)$
  - Same weight for each cell
- **Dynamic simulation**
  - Calculate the number of transitions, instead of power consumption

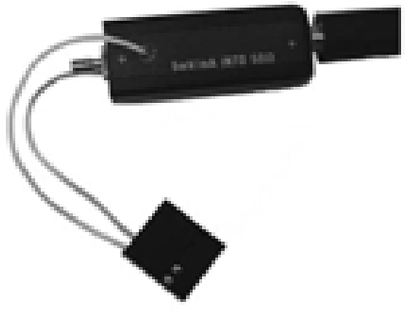
Power Models for Attackers

- **Attackers have limited information**
- **Behavioral simulation**
  - Hamming-distance model
    - For available partial (logical) netlist
    - Good for simulating data bus
  - Hamming-weight model
    - No netlist information at all
    - Related only to the data value (how many 1's), not the transition
    - Though not accurate estimation, but still may correlate with the power consumption

Comparisons of HW and HD Power Models

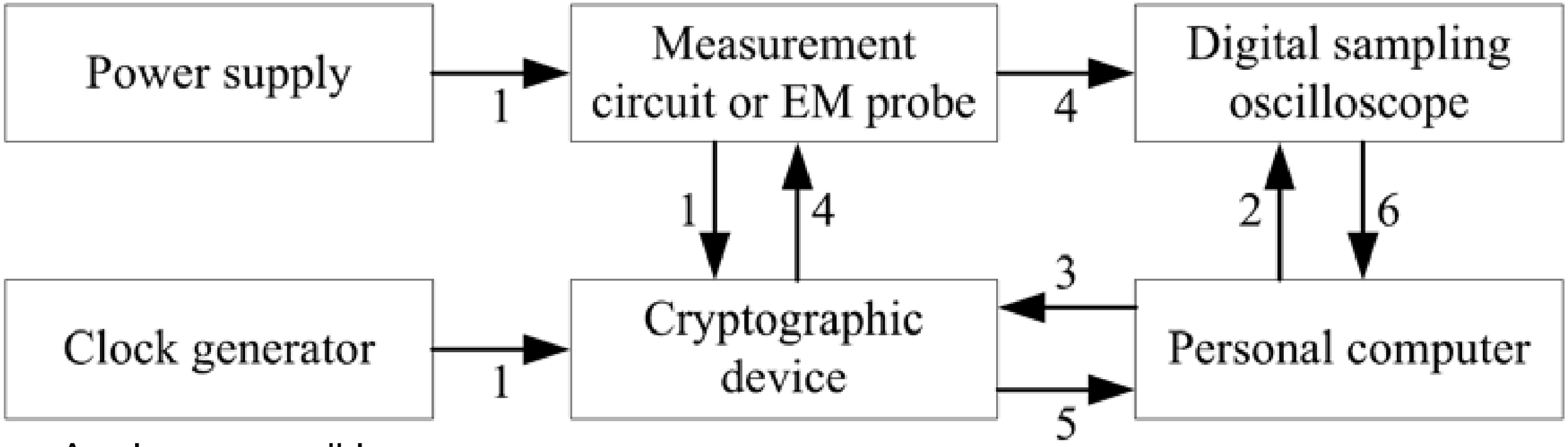
Simulation level	Behavioral	Behavioral
Power model	HD power model	HW power model
Required knowledge about the circuit	Processed data and “guess” about architecture	Processed data
Resource usage	Very low	Very low
Signal transitions considered	Equal for all bits	Equal for all bits
Glitches considered	No	No
Simulation result	Number of transitions	Number of logic ones

Typical Setup for Power Measurement



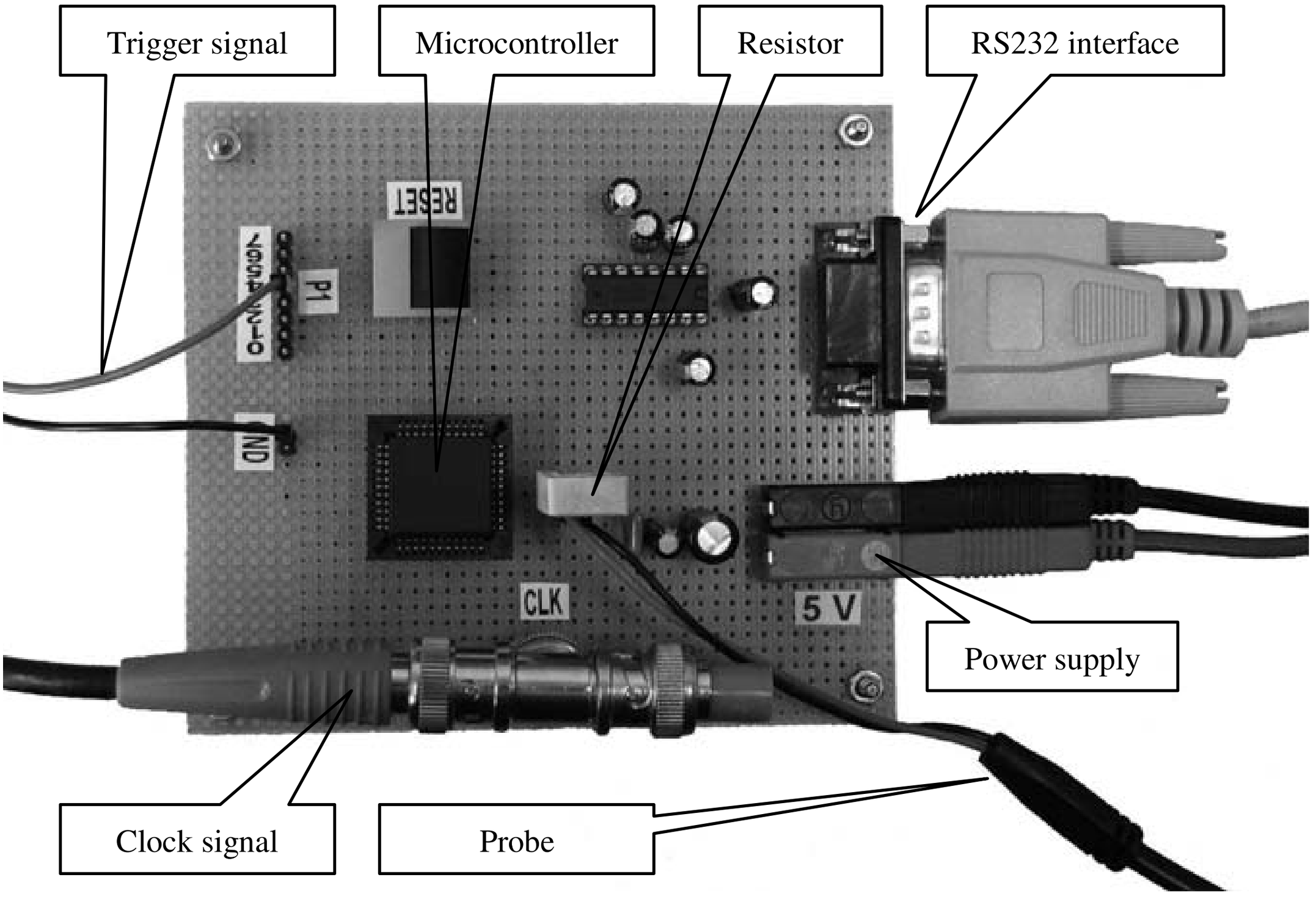




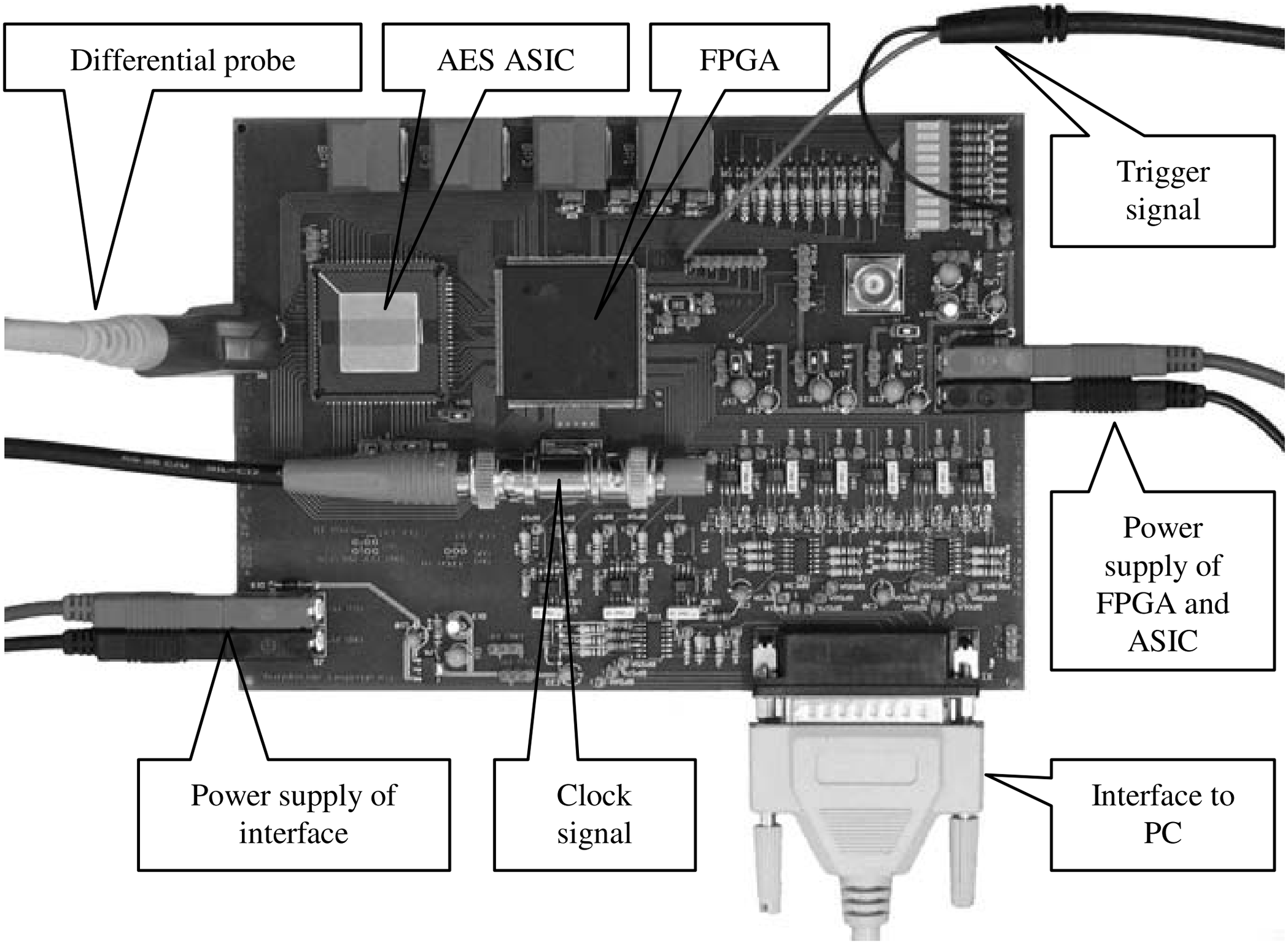


As slow as possible,
e.g. 2MHz

Example: 8-bit Microcontroller

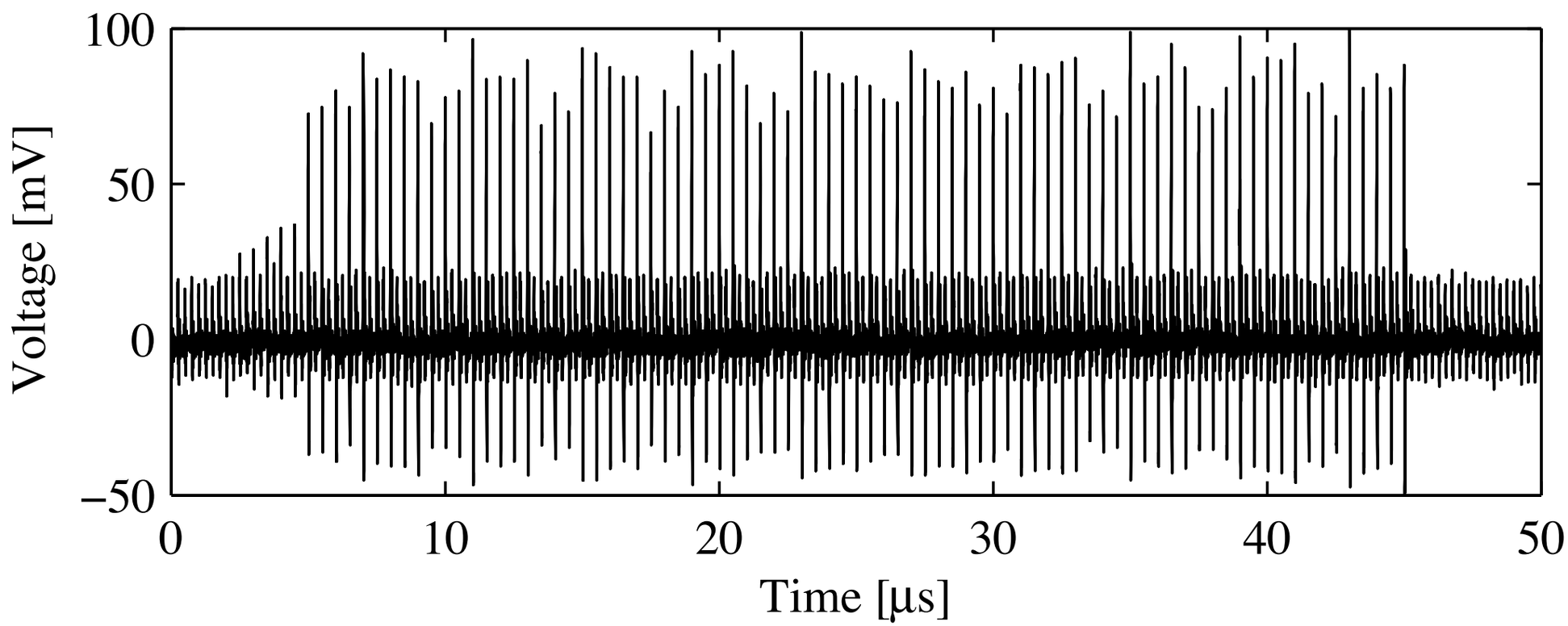


Example: AES ASIC



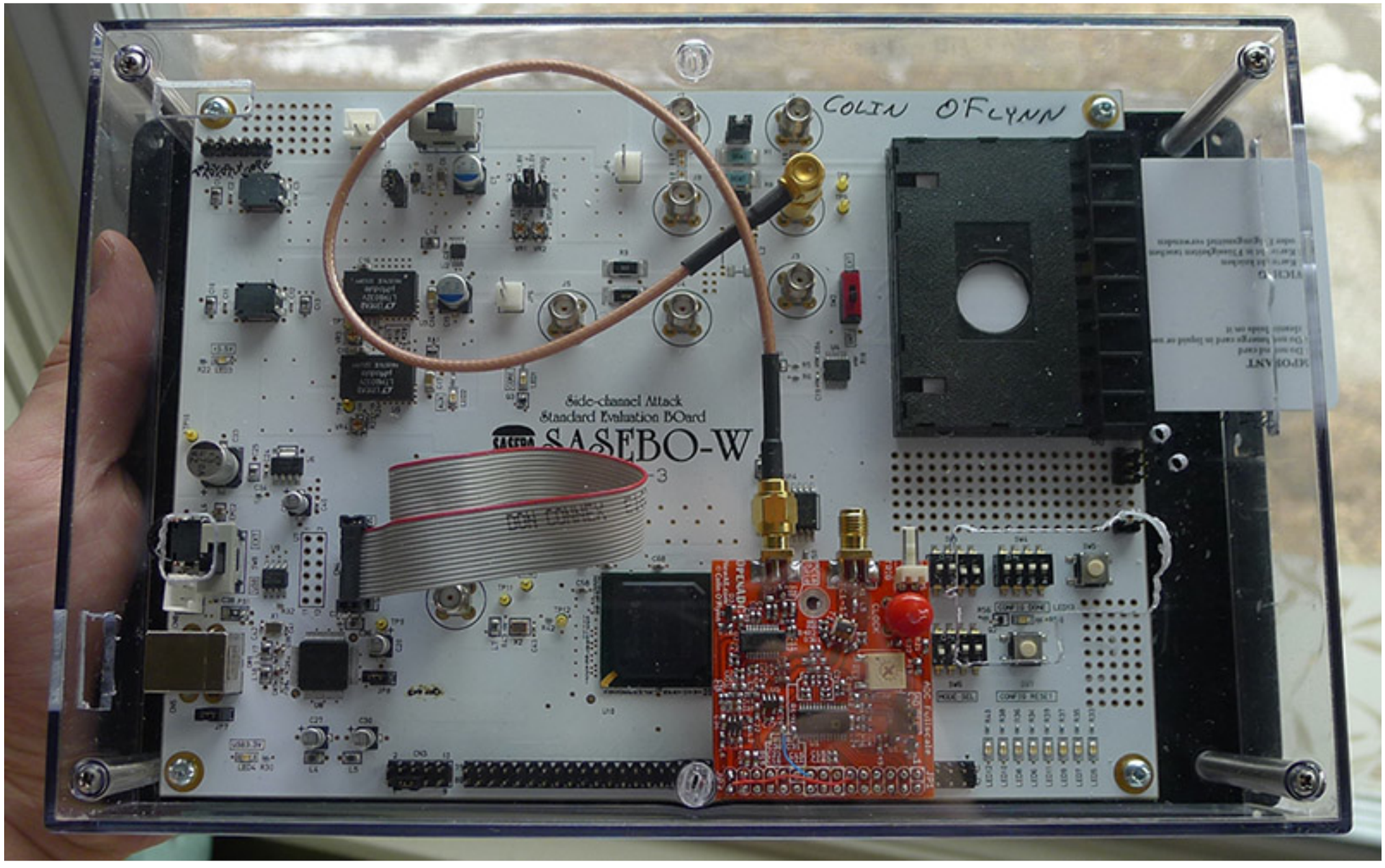
Example: AES ASIC

- **Measured waveform**
  - 10 AES encryption rounds, each taking 8 cycles
  - One positive peak for one active cycle



Example: Smart Card

SASEBO-W (Side-channel Attack Standard Evaluation BOard)



http://www.colinoflynn.com/tiki-index.php?page=SASEBOW

Quality of the Measurement

- **Electronic noise**
  - From other electronic devices
- **Switching noise**
  - From regions that we are not interested in
    - Switching induces large current
    - Parallelism increases the difficulty of attacks
- Filtering due to power ring (C) and bonding wire (L)
  - Unable to measure the original high-frequency current signal
  - Only able to measure filtered current

Influence of the Clock Frequency

- Interferences among clock cycles
- The attack may still succeed even when there is a large amount of noise.
- More noise -> more difficult to extract the secrets.

